Circular Equal-Mass Hard Rods after Rotation Reduction: Exact Free-Flow Conjugacy, All Events, and Return Times

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Abstract

For N indistinguishable equal-mass rods of length a on a circle of circumference $\ell > Na$, we identify the exact dynamical object on which the usual excluded-volume compression is globally valid. The conjugacy resolves binary, disjoint simultaneous, and genuine multiple collisions, preserves momentum and energy, and proves an explicit compact-interval no-Zeno bound. It also exposes a necessary boundary: the unreduced physical angle is an omitted cocycle, as the one-rod return period already demonstrates.

1 The corrected phase space

The frozen release obstruction identifier is `HEN-0280`. Write $\mathbb{T}_r = \mathbb{R}/r\mathbb{Z}$ and orient it clockwise. A rod is represented by its left endpoint. Choose a cyclic lift $x_1, \ldots, x_N \in \mathbb{R}$ satisfying

$$x_{i+1} - x_i \geq a \quad (1 \leq i < N), \qquad x_1 + \ell - x_N \geq a. \tag{1}$$

The free gaps are the left sides of (1) minus a; they are nonnegative and sum to

$$L = \ell - Na > 0.$$

We quotient global constant rotations $x_i \mapsto x_i + r$ and cyclic changes of the first rod. On a contact face, incoming and outgoing velocity labels are glued by the equal-mass elastic reflection. At an intersection of faces, the gluing is generated by the adjacent transpositions in the maximal contact block. The resulting rotation-reduced, unlabeled, collision-glued phase space is denoted by $\mathcal{R}_{N,\ell,a}$.

The free target is

$$\mathcal{F}_{N,L} = \big((\mathbb{T}_L \times \mathbb{R})^N\big)\big/\big(S_N \times \mathbb{T}_L\big), \tag{2}$$

where S_N permutes position–velocity pairs and $c \in \mathbb{T}_L$ sends every y_i to $y_i + c$ but leaves every velocity unchanged. Thus common drift is retained as phase data even though its spatial motion disappears in shape space. No time-dependent Galilean quotient is taken.

2 Compression and complete-flow conjugacy

For a representative satisfying (1), define

$$y_i = x_i - (i-1)a, \qquad 1 \leq i \leq N, \tag{3}$$

and reduce the y_i modulo L in (2).

Lemma 1 (Cyclic-start identity)**.** *If the next rod is chosen as the first rod, then the compressed lifted list is*

$$(y_2 + a, \ldots, y_N + a, y_1 + L + a).$$

Consequently its class in (2) is unchanged.

Proof. The new lifted endpoint list is $(x_2, \ldots, x_N, x_1 + \ell)$. Subtracting $(i-1)a$ gives the displayed list, because $\ell - (N-1)a = L + a$. Modulo L this is a cyclic permutation followed by one common translation by a. A global rod rotation similarly becomes a common translation. Hence (3) is well defined precisely on the reduced source. $\square$

Theorem 1 (Exact shape conjugacy)**.** *Compression induces a bijection*

$$C : \mathcal{R}_{N,\ell,a} \longrightarrow \mathcal{F}_{N,L}.$$

For every $t \in \mathbb{R}$, it conjugates the complete hard-rod flow R_t to the free flow

$$F_t[(y_i, v_i)_{i=1}^N] = [(y_i + tv_i, v_i)_{i=1}^N], \qquad CR_t = F_tC. \tag{4}$$

Proof. In cyclic order, (3) converts the free rod gaps exactly into the cyclic gaps between the y_i; both are nonnegative and sum to L. Conversely, cyclically order any representative of the free points, choose lifts $y_1 \le \cdots \le y_N \le y_1 + L$, and put $x_i = y_i + (i-1)a$. Then (1) holds. Common translation changes only the removed rod rotation, a different first point gives the cyclic relation in Lemma 1, and a permutation changes no unlabeled state. Ties are exactly contact faces, where the collision gluing removes the remaining velocity-order ambiguity. This proves bijectivity.

Between contacts, subtracting constants in (3) does not change velocities, so both sides follow (4). At an ordinary equal-mass collision, the rods exchange velocities; letting the two free phase points cross produces the same unlabeled pair. Adjacent transpositions generate every permutation within a simultaneous contact block. Therefore the identity persists through every collision, and the free group defines both forward and backward hard-rod evolution without an event-order convention. □

Jepsen's equal-mass hard-rod dynamics and free-crossing mechanism are classical [1], as is the excluded-volume setting of the Tonks gas [2]. Our result is a normalized source-model closure and not a claim of literature priority.

3 All events, conservation, and no Zeno

Fix a free representative. For $i \neq j$ with $v_i \neq v_j$, its pair coincidences are exactly

$$E_{ij} = \{t \in \mathbb{R} : (y_i - y_j) + t(v_i - v_j) \in L\mathbb{Z}\}. \tag{5}$$

At one event time, group all indices with the same compressed position. A group with at least two distinct velocities is a maximal collision block; different positions give simultaneous disjoint blocks.

Theorem 2 (All-event law)**.** *Immediately before a collision block, its spatial velocity list is decreasing up to ties; immediately after, it is increasing. Thus the event replaces the incoming list by the same multiset in sorted order. Binary collision is velocity exchange, and a genuine multiple collision is independent of any binary resolution order in the collision-glued quotient.*

Every event conserves

$$P = \sum_{i=1}^N v_i, \qquad E = \frac{1}{2}\sum_{i=1}^N v_i^2.$$

For a horizon $H \ge 0$, each unequal-velocity pair has at most

$$\left\lfloor \frac{H|v_i - v_j|}{L} \right\rfloor + 2 \tag{6}$$

coincidences in $[0, H]$. Hence only finitely many aggregated events occur on every compact interval: the flow has no Zeno accumulation.

Proof. Near a common free position at time t_0, the local positions are $y_0 + (t - t_0)v_i$. For $t < t_0$, increasing spatial order therefore reads the velocities from large to small; for $t > t_0$, it reads them from small to large. Sorting is a permutation, so its first and second power sums, hence P and E, are unchanged.

Equation (5) is one affine congruence. As t ranges over $[0, H]$, its left side crosses an interval of length $H|v_i - v_j|$; the number of multiples of L in that interval is bounded by (6). Summing over the finitely many pairs proves local finiteness. If $v_i = v_j$, the two free points either never coincide or remain coincident; the latter is a persistent contact and not an isolated event. □

Initial contacts are allowed because the phase space glues both one-sided velocity assignments. The construction is reversible for all real time. The strict inequality $L > 0$ is indispensable; at close packing the target circle collapses.

4 Why the unreduced statement is false

Lemma 1 shows the obstruction directly: changing the cyclic starting rod translates all compressed points by a. A permutation quotient alone does not remove that translation. There is an even shorter dynamical counterexample. For $N = 1$ and $v \neq 0$, the full physical rod returns after

$$T_{\text{physical}} = \frac{\ell}{|v|}, \tag{7}$$

whereas a naive point on the free circle $L = \ell - a$ would return after $L/|v|$. After rotation reduction, the configurational shape is one point and every fixed-velocity phase state is an equilibrium, exactly as Theorem 1 says.

Thus the complete physical rotation angle is an additional cocycle. This paper neither reconstructs it nor claims a conjugacy of the unreduced phase space. The obstruction is part of the theorem boundary, not a removable choice of notation.

References

[1] D. W. Jepsen, "Dynamics of a Simple Many-Body System of Hard Rods," *Journal of Mathematical Physics* 6(3) (1965), 405–413, doi:10.1063/1.1704288.

[2] L. Tonks, "The Complete Equation of State of One, Two and Three-Dimensional Gases of Hard Elastic Spheres," *Physical Review* 50 (1936), 955–963, doi:10.1103/PhysRev.50.955.